



Zarban

Adaptive Math Assessment & Learning Platform

Client Handover Brief · Grades 5–10 · NCERT-aligned

Live: zarban.zarbanlabs-app.workers.dev

What it is

Zarban is a complete platform that assesses a student's math ability, **diagnoses the reason behind every mistake**, and turns that into targeted practice — with a full console for teachers and administrators to monitor and manage everything.

The problem it solves

Ordinary online tests give a **score**. A score says a student is weak; it never says *why* or *what to do next*. A 60% could be a calculation slip, a reading-comprehension gap, or a missing skill from two grades below — and each needs a completely different fix.

Zarban finds the **cause**. It adapts to each student as they answer, and produces a report that reads like a tutor's assessment: "strong at the maths, but word problems are slowing them down — that's a reading gap, not a maths gap."

Who uses it

Students

Take an adaptive test (no marks shown during it), get a clear diagnostic report, and practise the exact gaps.

Teachers

Monitor classrooms and cohorts, spot the skills blocking the most learners, and drill into any student.

Administrators

Manage the question bank, curriculum, staff, system health, and the entire database.

Find out exactly where your **math stands**.

A smart diagnostic for Grades 5–10, aligned to the NCERT syllabus. Around 20–30 questions, about 15 minutes — and a personal report that shows your strengths, your gaps, and what to study next.



Adapts as you answer

Questions get easier or harder based on how you're doing — no two tests are the same.



Finds the why, not just the what

Every wrong option is mapped to a specific misconception, so the report explains your mistakes.



A report worth reading

Skill mastery, five learning dimensions, and the exact foundations to strengthen next.



Start your assessment

No marks shown during the test — just answer honestly.

Your name

e.g. Aarav Sharma

School (optional)

e.g. DAV Public School

Your class

5

CLASS

6

CLASS

7

CLASS

8

CLASS

9

CLASS

10

CLASS

Start Assessment

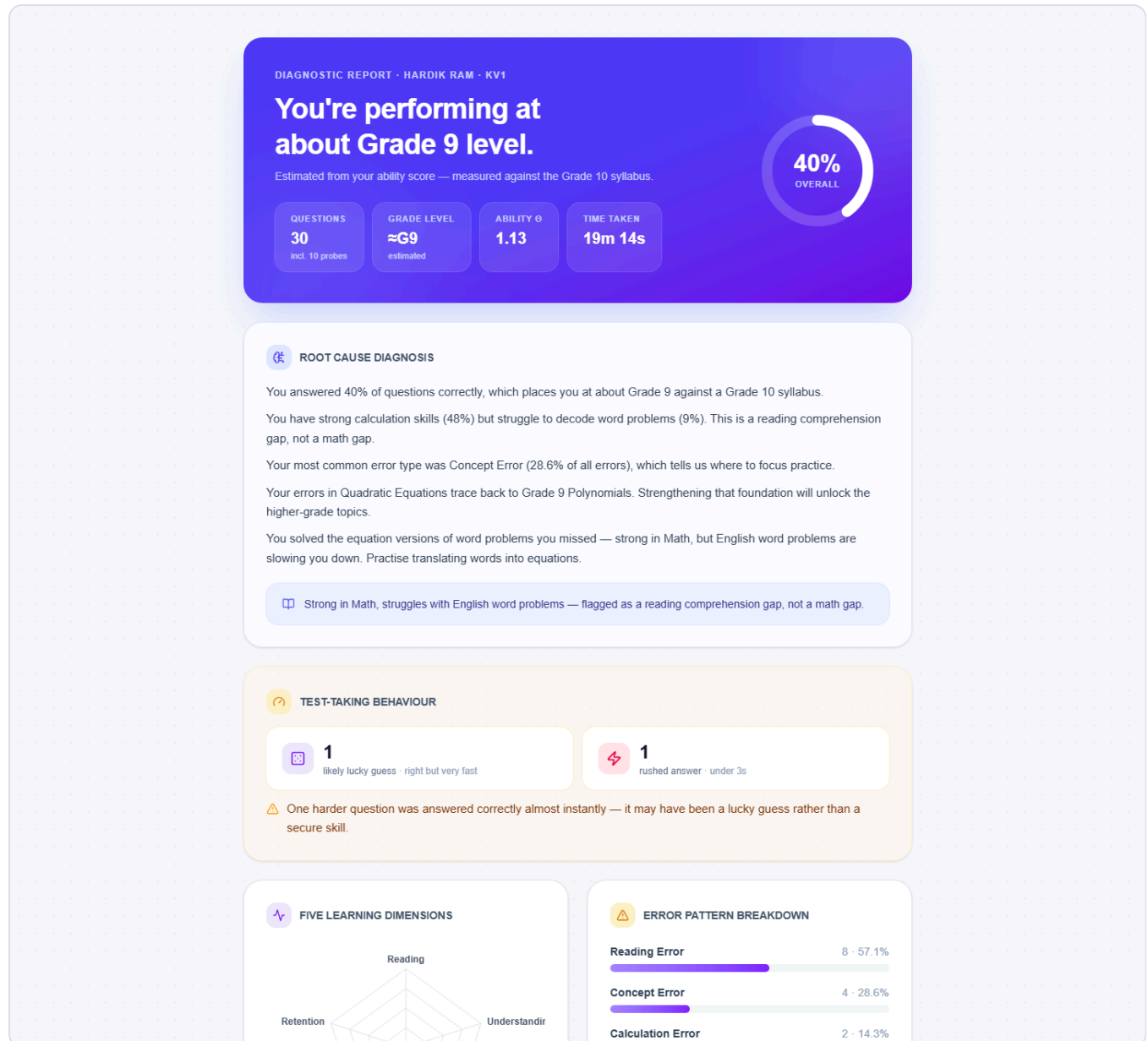
Powered by CAT · IRT · CDM · BKT diagnostics

Teachers: [open the admin dashboard](#) →

The student landing page — a class is chosen and the adaptive assessment begins.

The student journey

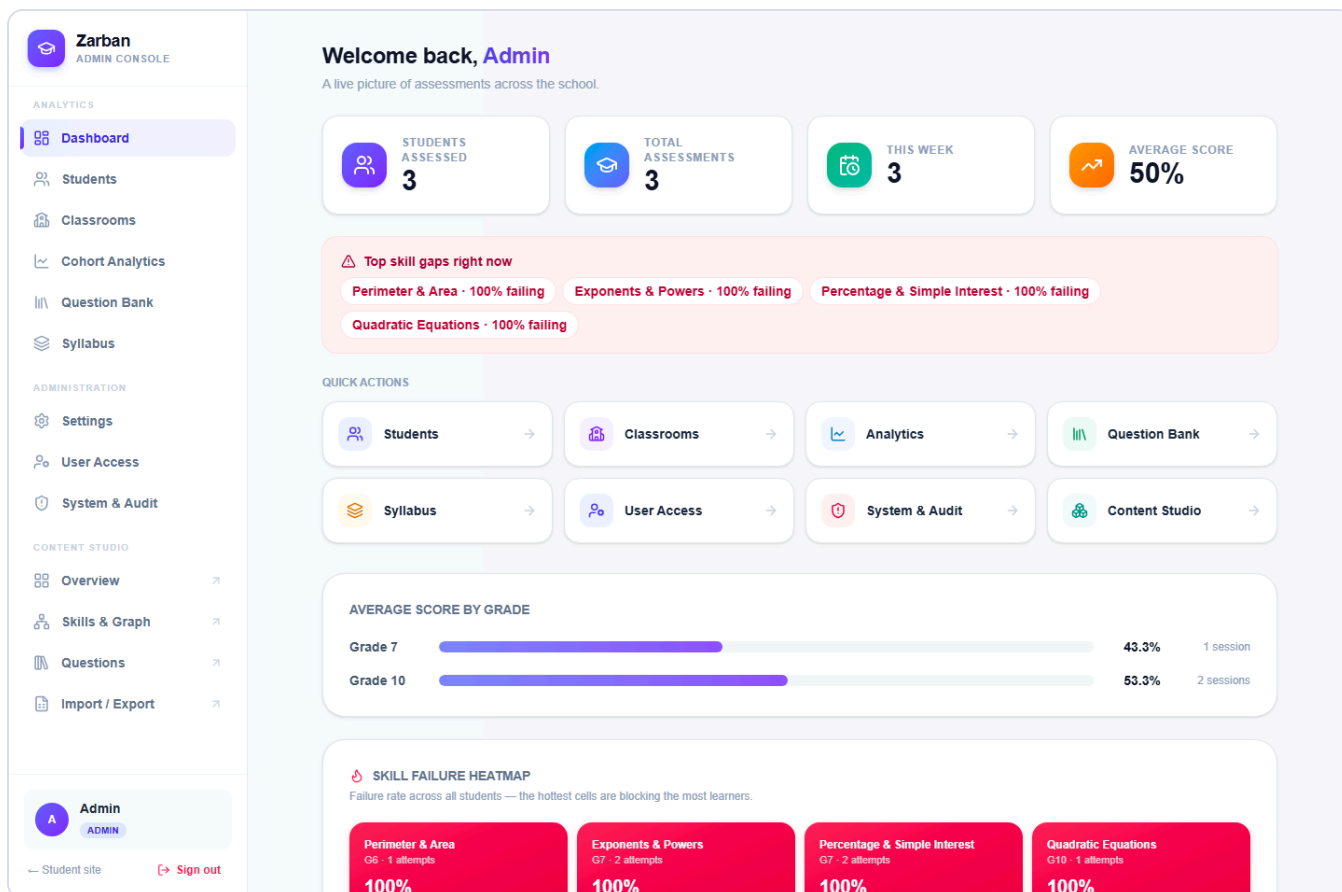
The test **adapts in real time** — get answers right and it gets harder; struggle and it steps down toward the foundational skills that are actually the problem. When it ends, the student receives a diagnostic report.



The diagnostic report: an estimated grade level, a plain-English root-cause diagnosis, flags for lucky guesses and rushed answers, a five-dimension learning profile, and a breakdown of error types.

The teacher & admin console

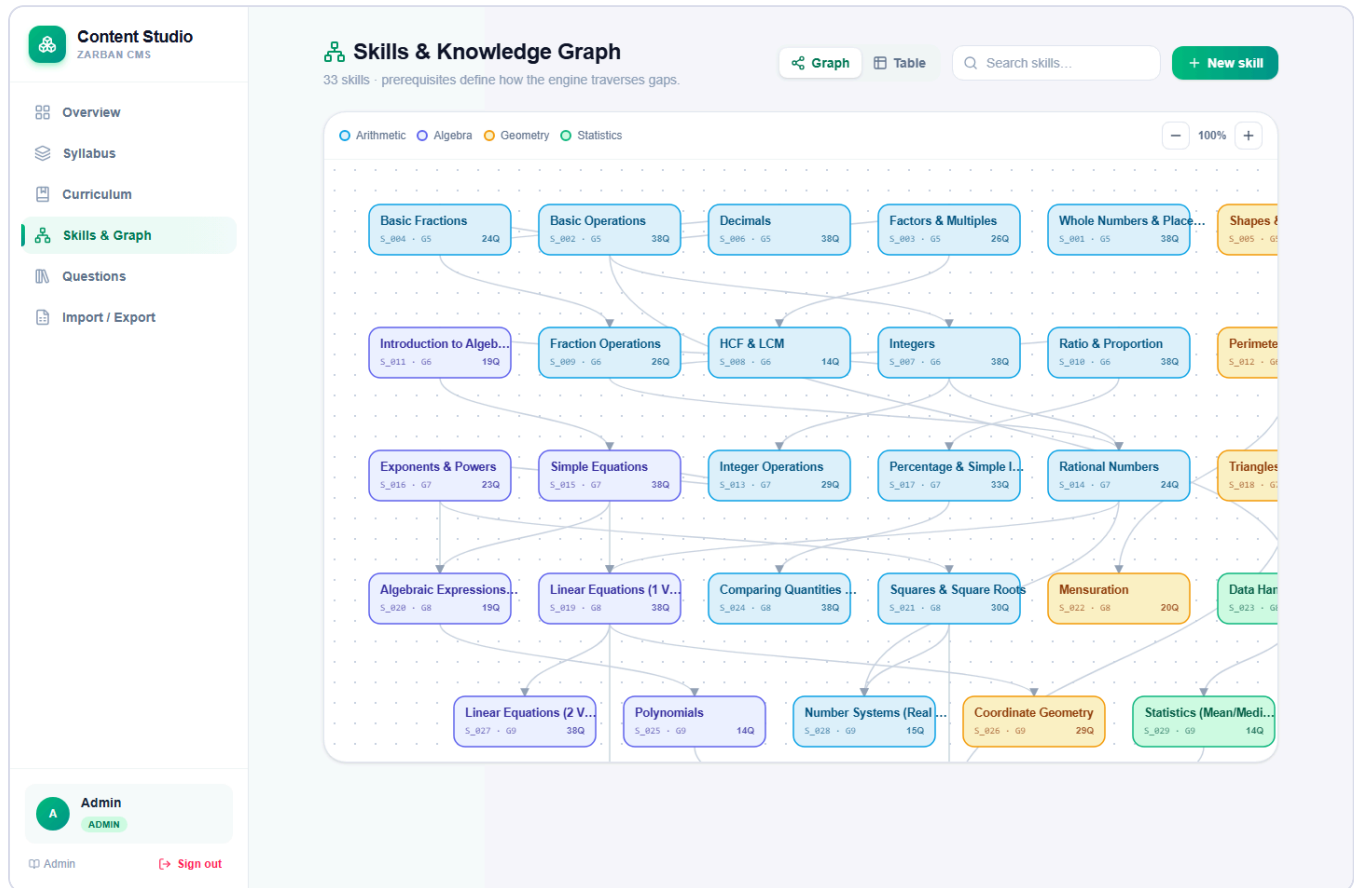
A single console gives staff a live picture of the whole school — headline numbers, the “hottest” skill gaps right now, average score by grade, and a class-level heatmap of where learners are failing.



The admin command center — totals, top skill gaps, quick actions, and the skill-failure heatmap.

The knowledge behind it

Every skill in the curriculum is mapped into a **knowledge graph** of prerequisites. That map is what lets the platform trace a Grade 10 mistake back to a missing Grade 9 skill and route the student there — the intelligence that makes the diagnosis meaningful.



The interactive skills knowledge graph — the prerequisite map the engine navigates.

Full control of the data

Administrators can browse and manage **every table** in the live database directly — searchable, editable, and safe (every change is recorded).

Zarban

ADMIN CONSOLE

ANALYTICS

Dashboard

Students

Classrooms

Cohort Analytics

Question Bank

Syllabus

ADMINISTRATION

Settings

User Access

Database

System & Audit

CONTENT STUDIO

Overview

Skills & Graph

Questions

Import / Export

Admin

ADMIN

← Student site

→ Sign out

Database

Browse and manage every table in the live database. Changes are immediate and audited.

TABLES · 29

admin_audit_log13

admin_users4

answer_traps2,733

bkt_state23

classrooms0

dimension_scores2

knowledge_graph36

q_matrix1,500

question_dimensions911

questions911

responses60

review_flags0

sessions3

settings2

skills33

students3

questions

911 rows · 17 columns

QUESTION_IDPKTEXT

QUESTION_TEXTTEXT

QUESTION_TYPETEXT

W

ncrt_10_ap_0000

What is the common difference of the AP...

MCQ

0

ncrt_10_ap_0001

What is the common difference of the AP...

MCQ

0

ncrt_10_ap_0002

What is the common difference of the AP...

MCQ

0

ncrt_10_ap_0003

What is the common difference of the AP...

MCQ

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ncrt_10_ap_0004

What is the common difference of the AP...

MCQ

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ncrt_10_ap_0005

What is the common difference of the AP...

MCQ

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ncrt_10_ap_0006

What is the common difference of the AP...

MCQ

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ncrt_10_ap_0007

What is the common difference of the AP...

MCQ

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ncrt_10_ap_0008

What is the common difference of the AP...

MCQ

0

ncrt_10_ap_0009

Find the 11th term of the AP 6, 11, 16,...

MCQ

0

ncrt_10_ap_0010

Find the 10th term of the AP 5, 7, 9, ...

MCQ

0

ncrt_10_ap_0011

Find the 15th term of the AP 4, 6, 8, ...

MCQ

0

ncrt_10_ap_0012

Find the 11th term of the AP 4, 6, 8, ...

MCQ

0

ncrt_10_ap_0013

Find the 9th term of the AP 8, 11, 14, ...

MCQ

0

ncrt_10_ap_0014

Find the 13th term of the AP 5, 11, 17,...

MCQ

0

ncrt_10_ap_0015

Find the 8th term of the AP 3, 5, 7, ...

MCQ

0

ncrt_10_ap_0016

Find the 13th term of the AP 3, 7, 11, ...

MCQ

0

The database manager — every table, searchable and editable, from inside the console.

Results & value

- **Comprehensive content:** 911 quality-checked questions across Grades 5–10, each mapped to a skill and its common misconceptions.
- **Genuinely adaptive:** five assessment techniques working together (item-response theory, knowledge tracing, misconception modelling, and a reading-vs-maths “twin” probe).
- **Fast and responsive:** the platform was performance-tuned end to end.
- **Zero hosting cost:** runs live on Cloudflare's free tier — a school can pilot it at no infrastructure cost.
- **Reliable:** 136 automated end-to-end checks guard every release.

Measured speed improvements

Interaction	Before	After	Faster by
Answering a question	~480–610 ms	~280–340 ms	~45%
Loading the dashboard	~311 ms	~170 ms	~45%
Loading the content hub	~313 ms	~150 ms	~52%

What's included in the handover

- **Live application** — zarban.zarbanlabs-app.workers.dev, deployed and running.
- **Full source code** — the complete repository, documented.
- **Interactive product tour** and a **technical deep-dive document** covering every workflow.
- **Admin access** — demo sign-in **admin@zarban.local** / **admin123** (change on first use, under User Access).

In one line: Zarban turns a raw test score into an actionable diagnosis and the exact next step — for the student, the teacher, and the administrator — and it's live today.